

SMEsPlus SaaS Foundation — Repository Update Pack v0.5

สถานะ

Approved for documentation completion.

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1. FDS v0.2 Completion

Foundation Domains

Tenant Management

- สร้าง tenant
- แก้ไข tenant profile
- suspend/reactivate tenant
- assign plan
- ตรวจสอบ usage quota
- ตรวจสอบ tenant health

Acceptance Criteria:

- Tenant code ต้อง unique
- Suspend tenant แล้ว user ภายใน tenant ต้อง login ไม่ได้
- ทุก critical action ต้องบันทึก audit log

Company / Branch Management

- จัดการ company
- จัดการ branch
- จัดการ division
- รองรับ tax ID และข้อมูลนิติบุคคล
- รองรับ organization hierarchy

Acceptance Criteria:

- Company ต้องอยู่ภายใต้ tenant เท่านั้น
- Branch ต้องผูกกับ company
- ห้าม query ข้าม tenant

IAM

- Login / refresh token
- Invite user
- Assign role
- MFA status
- Deactivate user
- Permission audit

Acceptance Criteria:

- Backend ต้อง enforce permission
- Frontend ใช้เพื่อซ่อน UI เท่านั้น ไม่ใช่ security control หลัก
- User email unique ภายใน tenant

Subscription & Module Activation

- Plan catalog
- Module catalog
- Plan-module mapping
- Tenant module activation
- Dependency check
- Billing impact preview

Acceptance Criteria:

- Module ที่ plan ไม่รองรับต้อง activate ไม่ได้
- Module dependency ต้องผ่านก่อน activation
- Activation ต้องมี audit log

Approval Foundation

- Approval workflow
- Approval request
- Approve/reject/delegate
- Approval timeline
- Escalation baseline

Acceptance Criteria:

- Reject ต้องระบุเหตุผล
- Approval action ต้อง immutable
- Requester ต้องเห็นสถานะล่าสุด

Notification Foundation

- Notification list
- Read/unread
- Approval notification
- System notification

Acceptance Criteria:

- Notification ต้อง scoped ตาม tenant/user
- ผู้ใช้เห็นเฉพาะ notification ของตนเอง

Integration Foundation

- API client
- API key
- Webhook
- Event log
- Retry failed webhook

Acceptance Criteria:

- API secret แสดงครั้งเดียว
- Webhook ต้องมี signing secret
- Failed event ต้อง retry ได้

Audit & Governance

- Audit search
- Filter by user/action/resource/date
- Export
- Immutable audit trail

Acceptance Criteria:

- Audit log ห้ามแก้ไข/ลบ
- Critical action ต้องมี request_id, actor_id, tenant_id, IP
- Export จำกัดเฉพาะ authorized role

2. Document Status Normalization

เอกสาร architecture เดิมให้เปลี่ยนสถานะจาก Draft / Pending Review เป็น:

- Status: Approved for SDS Expansion
- Architecture Gate: Passed with Execution-Level Follow-up Required
- Baseline Version: v0.5
- Approved Scope: SaaS Foundation
- Not Approved Yet: Full Accounting, Inventory, HR, CRM production modules

3. Execution-Level Documents

SDS_FOUNDATION_v0.1.md

Core Services:

- Auth Service
- Tenant Service
- Company Service
- Branch Service
- User Service
- Role Permission Service
- Subscription Service
- Module Activation Service
- Approval Service
- Notification Service
- Audit Service
- Integration Service
- Report Service

Request Flow: 1. API Gateway receives request 2. Auth middleware validates JWT/API key 3. Tenant context resolver resolves tenant_id 4. Tenant status validator checks active/suspended status 5. RBAC/ABAC permission engine validates action 6. Repository applies tenant filter 7. Database RLS enforces tenant isolation 8. Business logic executes transaction 9. Audit log is written 10. Response returned with request_id

OPENAPI_FOUNDATION_v0.1.yaml

Required endpoint groups:

- /auth/login
- /auth/refresh
- /tenants
- /companies
- /branches
- /users
- /roles
- /permissions
- /modules
- /tenant-modules/activate
- /approval-requests
- /notifications
- /audit-logs
- /api-clients
- /webhooks
- /integration-events
- /reports

Standard Response:

```
{
  "success": true,
  "data": {},
  "meta": {
    "request_id": "uuid",
    "tenant_id": "uuid",
    "timestamp": "ISO-8601"
  },
  "errors": []
}
```

ERD_FOUNDATION_v0.1.md

Core tables:

- tenants
- plans
- modules
- plan_modules
- companies
- branches
- divisions
- users
- roles
- permissions
- role_permissions
- user_role_assignments
- tenant_module_activations
- approval_workflows
- approval_requests
- notifications
- api_clients
- webhooks
- integration_events
- audit_logs

Mandatory fields for tenant-owned tables:

- id
- tenant_id
- created_at
- created_by
- updated_at
- updated_by
- deleted_at
- version

SCREEN_SPEC_FOUNDATION_v0.1.md

Screens: 1. Dashboard 2. Tenant Administration 3. Company & Branch Management 4. User & Role Management 5. Module Activation 6. Configuration Center 7. Approval Inbox 8. Integration Center 9. Audit & Governance 10. Reports

Each screen must include:

- Purpose
- User stories
- Components
- Actions
- Permission rules
- Empty state
- Error state
- Audit requirement

PERMISSION_MATRIX_v0.1.md

Roles:

- Platform Admin
- Tenant Owner
- Company Admin
- Branch Manager
- Department User
- Approver
- Auditor
- API Client

Permission rules:

- Platform Admin จัดการ tenant และ plan
- Tenant Owner จัดการ company, user, module
- Company Admin จัดการ company scope
- Branch Manager เห็นเฉพาะ branch scope
- Auditor read-only
- API Client จำกัดตาม scope และ token permission

MIGRATION_PLAN_v0.1.md

Migration sequence: 1. Create extension UUID 2. Create tenants 3. Create plans/modules 4. Create company hierarchy 5. Create IAM tables 6. Create subscription/module activation 7. Create approval/notification 8. Create integration tables 9. Create audit_logs 10. Enable RLS 11. Add indexes 12. Add seed permissions

TEST_STRATEGY_FOUNDATION_v0.1.md

Required tests:

- Unit test

- API contract test
- Integration test
- Tenant isolation test
- Permission test
- RLS test
- Security test
- Performance test
- Migration rollback test
- UAT test
- DR restore test

Critical rule:

- Tenant isolation failure = release blocker

UAT_SCENARIOS_FOUNDATION_v0.1.md

UAT flows: 1. Platform Admin creates tenant 2. Tenant Owner creates company and branch 3. Admin invites user 4. Admin assigns role 5. User logs in and sees scoped dashboard 6. Tenant Owner activates module 7. User submits approval request 8. Approver approves/rejects 9. Auditor checks audit log 10. Admin creates webhook 11. Failed webhook retries successfully 12. Report exported by authorized user

ADR_INDEX_AND_APPROVAL_STATUS.md

ADR list:

- ADR-0010 SaaS First Architecture — Approved
- ADR-0011 Multi-Tenant by Design — Approved
- ADR-0012 API First — Approved
- ADR-0013 PostgreSQL Schema-per-Tenant + RLS — Approved for Foundation
- ADR-0014 RBAC + ABAC + RLS — Approved
- ADR-0015 Audit by Design — Approved
- ADR-0016 Kubernetes Deployment — Approved for Target Architecture
- ADR-0017 Webhook/Event Integration — Approved for Foundation
- ADR-0018 Thai Localization Foundation — Approved
- ADR-0019 Modular Monolith for MVP — Approved

DEVELOPMENT_READINESS_CHECKLIST.md

Ready:

- Architecture direction
- SaaS foundation
- Multi-tenant model
- Security baseline
- Deployment target
- Core domains
- Core screens
- Permission model
- Test strategy
- UAT baseline

Still required before production:

- Full OpenAPI request/response schemas
- Actual migration SQL files
- UI design system
- Complete ERD diagram
- CI/CD implementation
- Security automated test suite
- Load test report
- DR drill result

Final Status

SMEsPlus SaaS Foundation is now:

READY FOR:

- SDS review
- API mock
- DB migration drafting
- UX wireframe
- Sprint backlog creation
- Backend scaffold

NOT YET READY FOR:

- Production launch
- Full ERP module implementation
- Customer onboarding